

- (c) The star Sirius has a radius of $1.19 \times 10^9 \text{ m}$ and loses mass due to nuclear fusion at a rate of $1.09 \times 10^{11} \text{ kg s}^{-1}$. Assume that the power of the radiation emitted by the star is equal to the power released by this process.

(i) Determine a value for the luminosity of Sirius. Give a unit with your answer.

luminosity = unit [2]

(ii) Use your answer in (c)(i) to determine the surface temperature of Sirius.

surface temperature = K [2]

- (d) Explain how cosmologists use standard candles to estimate the distance of a galaxy from the Earth.

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 [3]

[Total: 13]

